

User Manual

Introduction

This user manual provides step-by-step instructions for using the Jenkins Continuous Integration (CI) system implemented in this project. It explains how to access Jenkins, run jobs, view build results, and manage pipelines.

Accessing Jenkins

Steps:

1. Open a web browser.
2. Enter the Jenkins URL:

```
http://localhost:8080
```

3. Log in using the administrator credentials (Enrolment Number as username).

Sign in to Jenkins

Username

23FE10CSE00309

Password

.....

☒ Keep me signed in

Sign in

Figure 1. Jenkins login page

Dashboard Navigation

After logging in, the Jenkins dashboard is displayed.

Main features of the dashboard:

- **New Item:** Create new jobs
- **Build History:** View previous builds
- **Manage Jenkins:** Configure system settings
- **Job List:** Displays all configured jobs

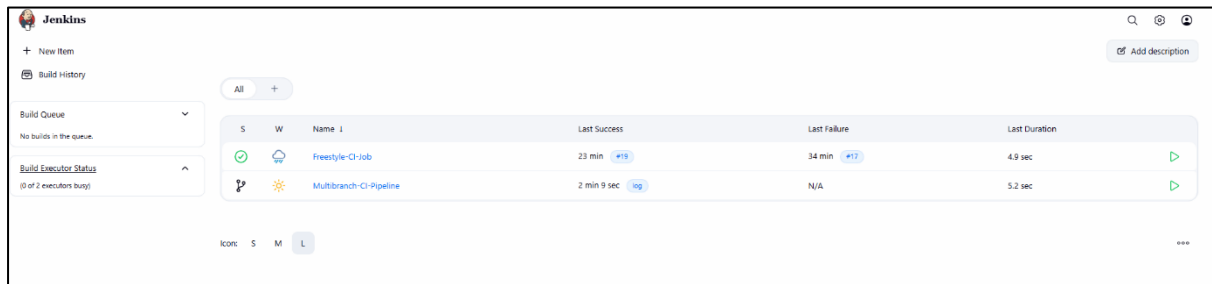


Figure 2. Jenkins dashboard with job list

Running a Freestyle Job

Steps:

1. Click on the Freestyle job name.
2. Click **Build Now**.
3. Jenkins starts executing the job.
4. Click on the build number to view details.

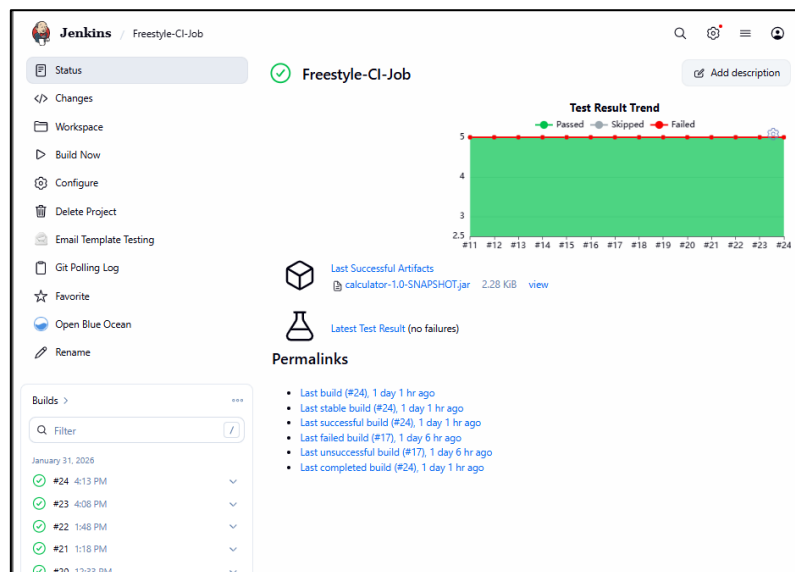


Figure 3. Freestyle job page with “Build Now” option

Viewing Build Console Output

Steps:

1. Click on the build number (e.g., #1).
2. Click on **Console Output**.

3. Observe the logs for:
 - a. Code checkout
 - b. Build execution
 - c. Test results

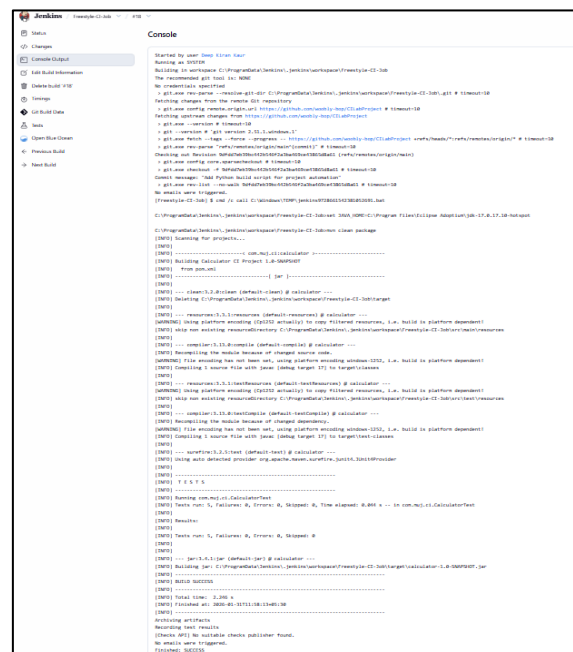


Figure 4. Console Output showing successful build

Running Multibranch Pipeline Job

Steps:

1. Click on the Multibranch Pipeline job.
2. Jenkins automatically displays discovered branches.
3. Click any branch to view its pipeline.
4. Click **Build Now** to trigger a pipeline run.

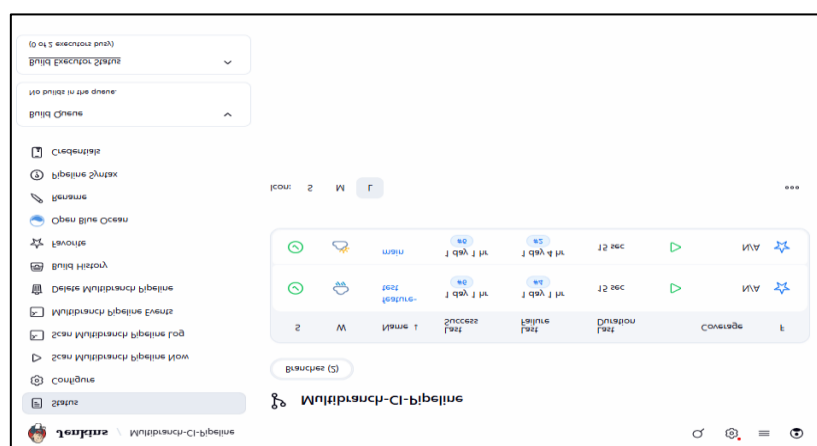
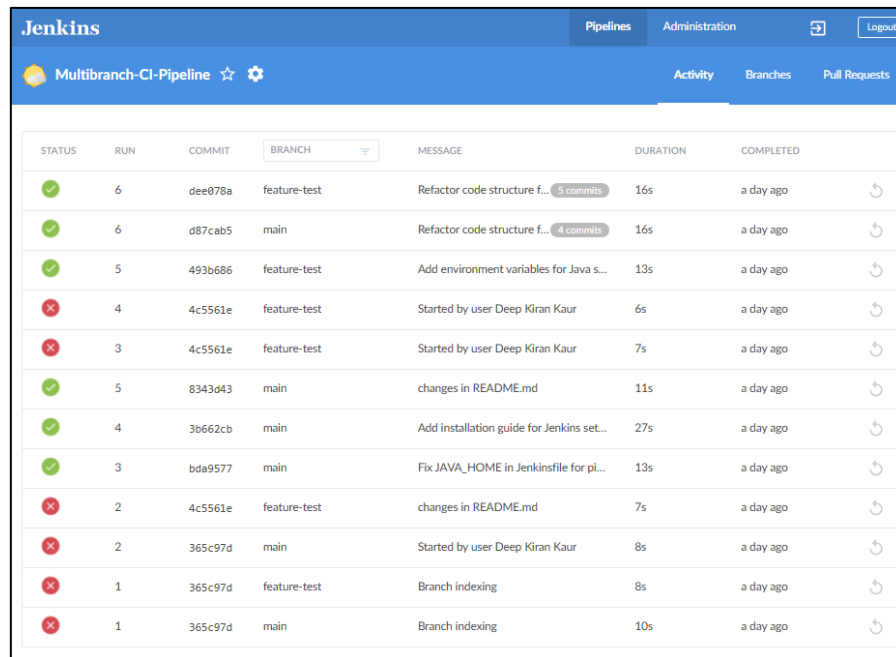


Figure 5. Multibranch Pipeline job showing branches

Viewing Pipeline Stages

Steps:

1. Click on a pipeline build.
2. View the stage-wise execution (Checkout, Build, Test, Deploy).
3. Click on any stage to view logs.



The screenshot shows the Jenkins interface for a pipeline named 'Multibranch-CI-Pipeline'. The top navigation bar includes 'Pipelines' and 'Administration'. Below the pipeline name, there are tabs for 'Activity', 'Branches', and 'Pull Requests'. The main table displays a list of pipeline runs with columns for STATUS, RUN, COMMIT, BRANCH, MESSAGE, DURATION, and COMPLETED. The table shows several runs, some successful (green checkmark) and some failed (red X). The most recent run (Run 6) is successful and shows a message 'Refactor code structure f... (5 commits)'.

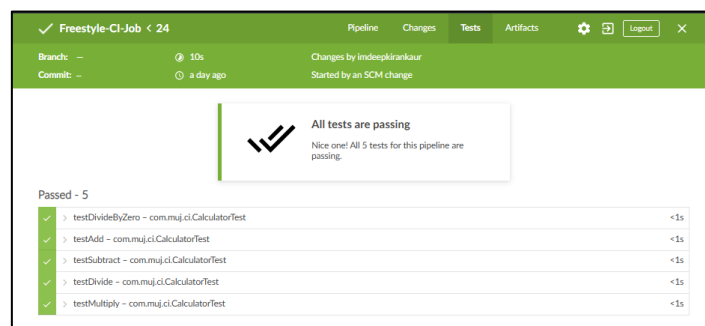
STATUS	RUN	COMMIT	BRANCH	MESSAGE	DURATION	COMPLETED
✓	6	dee978a	feature-test	Refactor code structure f... (5 commits)	16s	a day ago
✓	6	d87cab5	main	Refactor code structure f... (4 commits)	16s	a day ago
✓	5	493b686	feature-test	Add environment variables for Java s...	13s	a day ago
✗	4	4c5561e	feature-test	Started by user Deep Kiran Kaur	6s	a day ago
✗	3	4c5561e	feature-test	Started by user Deep Kiran Kaur	7s	a day ago
✓	5	8343d43	main	changes in README.md	11s	a day ago
✓	4	3b662cb	main	Add installation guide for Jenkins set...	27s	a day ago
✓	3	bda9577	main	Fix JAVA_HOME in Jenkinsfile for pi...	13s	a day ago
✗	2	4c5561e	feature-test	changes in README.md	7s	a day ago
✗	2	365c97d	main	Started by user Deep Kiran Kaur	8s	a day ago
✗	1	365c97d	feature-test	Branch indexing	8s	a day ago
✗	1	365c97d	main	Branch indexing	10s	a day ago

Figure 6. Pipeline stage view (Blue Ocean)

Viewing Test Results

Steps:

1. Open a completed build.
2. Click on **Test Result**.
3. View passed and failed test cases.



The screenshot shows the Jenkins Test Results page for a build named 'Freestyle-CI-Job < 24'. The top navigation bar includes 'Pipeline', 'Changes', 'Tests', and 'Artifacts'. The page shows a summary of test results with a green checkmark and the text 'All tests are passing'. Below this, there is a list of passed tests with their durations.

Test Name	Duration
testDivideByZero - com.muj.ci.CalculatorTest	<1s
testAdd - com.muj.ci.CalculatorTest	<1s
testSubtract - com.muj.ci.CalculatorTest	<1s
testDivide - com.muj.ci.CalculatorTest	<1s
testMultiply - com.muj.ci.CalculatorTest	<1s

Figure 7. Test Results page

Downloading Build Artifacts

Steps:

1. Open a completed build.
2. Click on **Artifacts**.
3. Download generated files.

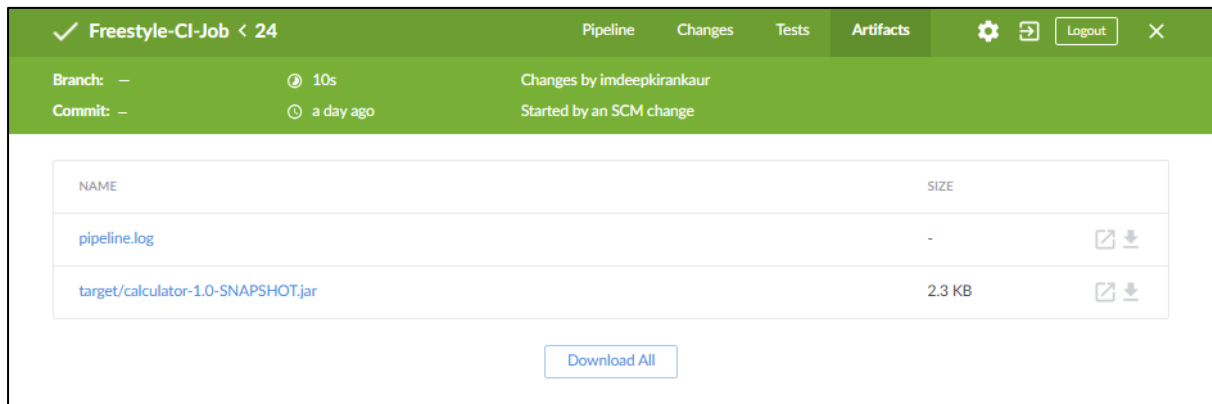


Figure 8. Artifacts section

Triggering Builds Automatically

Builds can be triggered automatically by:

- Pushing code to GitHub (via webhook)
- Polling SCM schedule
- Branch indexing in Multibranch Pipeline

This ensures Continuous Integration without manual intervention.

Logging Out

Steps:

1. Click on the username at the top-right corner.
2. Click **Log out**.

Summary

This user manual enables users to effectively operate the Jenkins CI system.

It covers login, job execution, pipeline monitoring, and result analysis.

Using these steps, users can continuously integrate and test their code using Jenkins.